

Linux Fundaments Part1 Lab

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Linux Fundamentals Part 1

Embark on the journey of learning the fundamentals of Linux. Learn to run some of the first essential commands on an interactive terminal.

20 min 9,08,518

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Stuck on a question? I am here to help you with real-time guidance, personalized hints, and explanations. May 27, 2021

Learn the Linux Fundamentals - Part 1

Source: YouTube

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Welcome to the first part of the "Linux Fundamentals" room series. You're most likely using a Windows or Mac machine, both are different in visual design and how they operate. Just like Windows, iOS and MacOS, Linux is just another operating system and one of the most popular in the world powering smart cars, android devices, supercomputers, home appliances, enterprise servers, and more.

We'll be covering some of the history behind Linux and then eventually starting your journey of being a Linux-wizard! This room will have you:

- Running your very first commands in an interactive Linux machine in your browser
- Teaching you some essential commands used to interact with the file system
- Demonstrate how you can search for files and introduce shell operators

Answer the questions below

Let's get started!

 No answer needed

✓ Correct Answer

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Flavours of Linux

The name "Linux" is actually an umbrella term for multiple OS's that are based on UNIX (another operating system). Thanks to Linux being open-source, variants of Linux come in all shapes and sizes - suited best for what the system is being used for.

For example, Ubuntu & Debian are some of the more commonplace distributions of Linux because it is so extensible. I.e. you can run Ubuntu as a server (such as websites & web applications) or as a fully-fledged desktop. For this series, we're going to be using Ubuntu.

Note: Ubuntu Server can run on systems with only 512MB of RAM!

Similar to how you have different versions Windows (7, 8 and 10), there are many different versions/distributions of Linux.

Answer the questions below

Research: What year was the first release of a Linux operating system?

1991

✓ Correct Answer

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Interacting With Your First Linux Machine (In-Browser)

Task 3: Running Your First few Commands

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Active Machine Information

Title	IP Address	Expires	
linuxfundpt1	10.10.144.238	1h 58m 49s	? Add I host Terminate

This contains all of the information for the machine deployed in the room including the IP address and expiry timer - along with buttons to manage the machine. Remember to "Terminate" a machine once you are done with the room. More information on this can be found in the [tutorial](#) room.

For now, press "Start Machine" where you will be able to interact with your own Linux machine within your browser whilst following along with this room:

Update me...

No answer needed

Completed

Task 4

Running Your First few Commands

Task 5

Flags and Switches

Task 6

Interacting With the Filesystem

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

tryhackme@ip-10-10-234-208:~\$

Answer the questions below

I've deployed my first Linux machine!

No answer needed

✓ Correct Answer

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Running Your First few Commands

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As shown in the terminal above, if we want to "echo" a single word, we don't need to use double quotes, for example, `echo Hello`. However, the string should be enclosed within double quotes if one or more spaces are present, for example, `echo "Hello Friend!"`.

`whoami` can be used to find the username we are logged in as.

Using whoami to find out the username of who we're logged in as

```
tryhackme@linux1:~$ whoami
```

Try this on your [Linux](#) machine now!

Answer the questions below

If we wanted to output the text "**TryHackMe**", what would our command be?

✓ Correct Answer [Hint](#)

What is the username of who you're logged in as on your deployed Linux machine?

✓ Correct Answer [Hint](#)

 Task 5 ✓ Interacting With the Filesystem!

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2. I have used the **"pwd"** (print working directory) command to find the full file path of this "Documents" folder.

3. We're helpfully told by Linux that this "Documents" directory is stored at `/home/ubuntu/Documents` on the machine — great to know!

4. Now in the future, if we find ourselves in a different location, we can just use `cd /home/ubuntu/Documents` to change our working directory to this "Documents" directory.

Answer the questions below

On the Linux machine that you deploy, how many folders are there?

✓ Correct Answer

Which directory contains a file?

✓ Correct Answer [Hint](#)

What is the contents of this file?

✓ Correct Answer

Use the `cd` command to navigate to this file and find out the new current working directory. What is the path?

✓ Correct Answer



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Using "grep" to find any entries with the IP address of "81.143.211.90" in "access.log"

```
tryhackme@linux1:~$ grep "81.143.211.90" access.log
81.143.211.90 - - [25/Mar/2021:11:17 + 0000] "GET / HTTP/1.1" 200 417 "-" "Mozilla/5.0 (Linux; Android 7.0; Moto G(4))"
tryhackme@linux1:~$
```

"Grep" has searched through this file and has shown us any entries of what we've provided and that is contained within this log file for the IP.

Answer the questions below

Use grep on "access.log" to find the flag that has a prefix of "THM". What is the flag? **Note:** The "access.log" file is located in the "/home/tryhackme/" directory.

THM{ACCESS}

✓ Correct Answer

Hint

And I still haven't found what I'm looking for!

No answer needed

✓ Correct Answer

Task 7 ✓ An Introduction to Shell Operators

Task 8 ✓ Conclusions & Summaries

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Answer the questions below

If we wanted to run a command in the background, what operator would we want to use?

&

✓ Correct Answer

If I wanted to replace the contents of a file named "passwords" with the word "password123", what would my command be?

echo password123 > passwords

✓ Correct Answer

Hint

Now if I wanted to add "tryhackme" to this file named "passwords" but also keep "passwords123", what would my command be

echo tryhackme >> passwords

✓ Correct Answer

Hint

Now use the deployed Linux machine to put these into practice

No answer needed

✓ Correct Answer

Task 8 ✓ Conclusions & Summaries

Task 9 ✓ Linux Fundamentals Part 2

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interact with a `Linux` machine.

I hope this room hasn't been too daunting for you to power-on through with. It's as I previously mentioned, you're going to become familiar with these things very quickly because of how often you're going to be using them.

To quickly recap, we've covered the following:

- Understanding why `Linux` is so commonplace today
- Interacting with your first-ever `Linux` machine!
- Ran some of the most fundamental commands
- Had an introduction to navigating around the filesystem & how we can use commands like `find` and `grep` to make finding data even more efficient!
- Power up your commands by learning about some of the important shell operators.

Take some time to have a play around in this room. When you feel a little bit more comfortable, progress onto [Linux Fundamentals Part 2](#)

Answer the questions below

I'll have a play around!

No answer needed

✓ Correct Answer

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Linux Fundamentals Part 2

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Task 6 ✓ Searching for Files

Task 7 ✓ An Introduction to Shell Operators

Task 8 ✓ Conclusions & Summaries

Task 9 ✓ Linux Fundamentals Part 2

Visit part two of the `Linux` fundamentals series here! <https://tryhackme.com/room/linuxfundamentalspart2>

Answer the questions below

Terminate the machine deployed in this room from task 3.

No answer needed

✓ Correct Answer

Join [Linux Fundamentals Part 2!](#)

No answer needed

✓ Correct Answer